Modeling, Identification and Control of Mechatronic Systems
FALL 2025

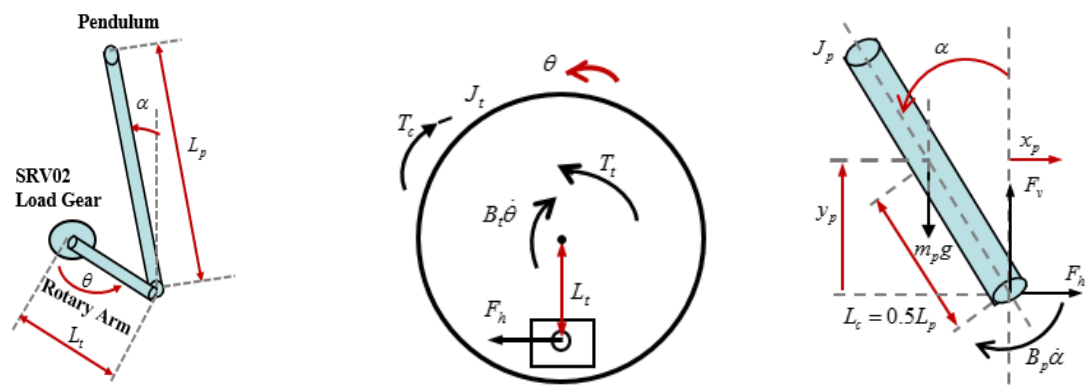
OUT: 09/23/25 DUE: 09/30/25

Use separate sheets to answer each problem with clear problem number, figures, equations and illustrations. Please **submit the answer sheets** together with **MATLAB codes and Simulink model** (MATLAB2022a Version or below) used in your problem analysis when you hand in your homework.

建模：列出元件方程，推导出系统数学模型，填写s-function和MATLAB代码，比较线性与非线性系统的仿真结果

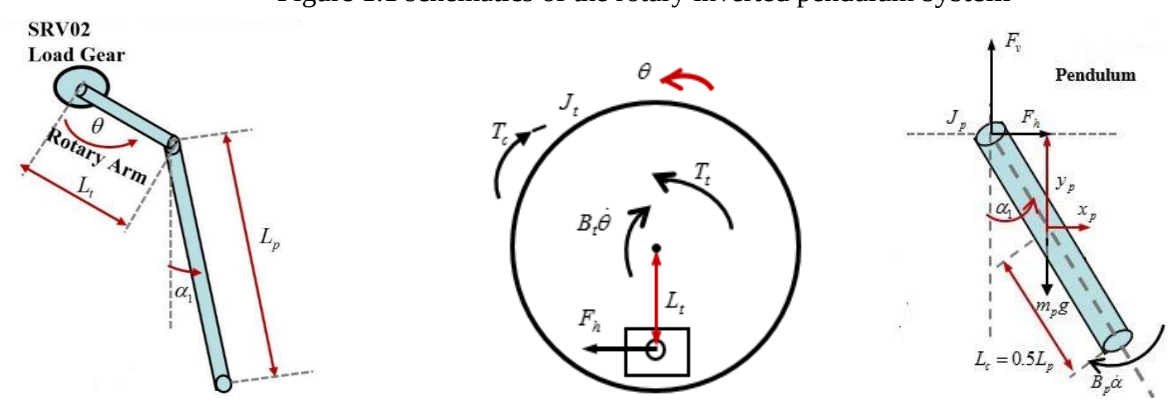
PROBLEM #1 (Modeling)

Assume a rotary inverted pendulum system shown in Figure 1.1, which consists of a flat arm, with a pivot at one end and a metal shaft on the other end. The pivot-end can be mounted on top of the SRV02 load gear shaft and fastened with screws. The actual pendulum link is fastened onto the metal shaft and the shaft is instrumented with a sensor to measure its angle. The result is a horizontally rotating arm with a pendulum at the end. The same system with the rod being downward (i.e., the rotary pendulum system) is shown in Figure 1.2. The SRV02 DC motor armature circuit and gear train is shown in Table 1.1.



(a) rotary inverted pendulum (b) FBD (free body diagram) of rotary arm (c) FBD of the rod

Figure 1.1 schematics of the rotary inverted pendulum system



(a) rotary pendulum (b) FBD (free body diagram) of rotary arm (c) FBD of the rod

Figure 1.2 schematics of the rotary pendulum system

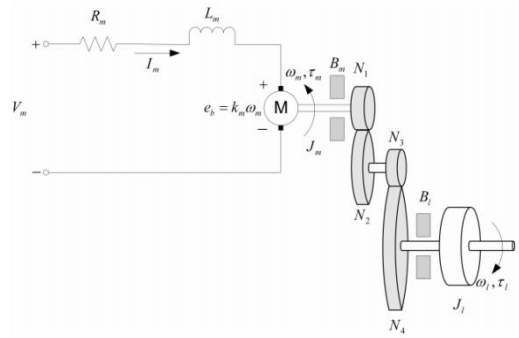


Figure 1.3 SRV02 DC motor armature circuit and gear train

Table 1.1 Rotary Pendulum system specifications

Symbol	Description	Value	Unit
m_p	Mass of pendulum	0.127	kg
L_c	Half of the total length of pendulum	0.1685	m
J_p	Pendulum moment of intertia about center of mass	0.0012	kg.m^2
B_p	viscous damping coefficient of the pendulum	0.0024	N.m.s/rad
L_r	Rotary arm length from pivot to tip	0.216	m
J_r	Rotary arm moment of inertia about its center of mass	0.001	kg.m^2
B_r	Rotary arm viscous damping coefficient as seen at the pivot axis	0.0024	N.m.s/rad
R_m	Motor armature resistance	2.6	Ω
k_t	Motor current-torque constant	7.68×10^{-3}	N.m/A
k_m	Motor back-EMF constant	7.68×10^{-3}	V/(rad/s)
K_g	Total gear ratio	70	
η_m	Motor efficiency	0.69	
η_g	Gearbox efficiency	0.9	
V_m	Applied voltage to the motor	-10~10	V
A_c	Coefficient of Coulomb friction torque	0.4	V

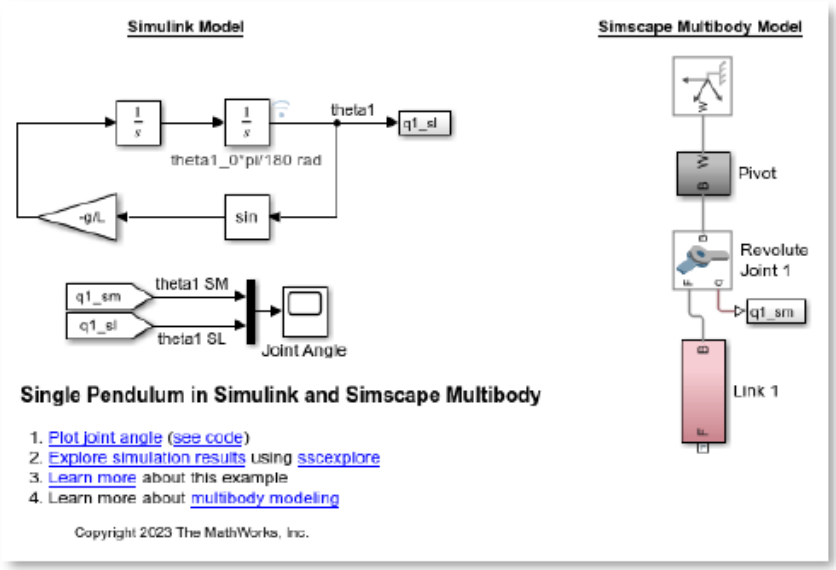
(A) According to the free-body diagram analysis, write down the **equations of motion (EOM)** (New-Euler Method) for the inverted pendulum system and the pendulum system respectively. (Please refer to lecture notes: Page 10 of MIC_L2)

(B) Derive the **nonlinear dynamics** and the **linearized dynamics** of the inverted pendulum system and the pendulum system respectively.

(C) According to the nonlinear and linear dynamics of the inverted pendulum system and the pendulum system respectively, fill in the key codes of establishing the **MATLAB/Simulink model for simulation**. Compare the **output response (rod angle, pendulum angle, control voltage)** of the nonlinear and linear pendulum/inverted-pendulum system using square wave input voltage with 0.8Hz and different amplitudes such as 0.8V/1.2V/1.6V/2V.

(D) (**Optional**) Establish the physical model of the rotary pendulum system using MATLAB Simulink and Simscape Multibody Tools. You can refer to the following example in MATLAB. And conduct simulation

with comparing the response of physical models and mathematical models you obtained in (C). Tell what results in the difference between physical model and mathematical model in simulation.



https://ww2.mathworks.cn/help/sm/ug/single-pendulum-in-simulink-and-simscape-multibody.html?s_tid=srchtitle_site_search_1_simscape+simulink

Figure 1.4